Report

Manchester Housing Price Dataset

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1. Introduction

This report presents an analysis of the Manchester housing dataset using Python programming. The main goal is to understand the housing market by looking at different variables like waterfront properties, floor space, and how the year a house was built relates to its price. I am working as a junior data scientist at Estate Management Plc, and this analysis will help predict future market trends.

The housing market is an important part of the UK economy. Understanding what makes house prices go up or down can help estate managers give better advice to their clients.

This report uses data analytics techniques to find patterns in the data and provide useful insights. The analysis covers data import, cleaning, statistical tests, and creating visualisations to show the findings.

This assignment helps me demonstrate my understanding of programming concepts and data manipulation techniques. I will explain the methods I used with references to support my decisions.

1.1 Features Discription

Bedrooms	Number of bedrooms in the house.
Bathrooms	Number of bathrooms in the property.
Living_area_internal (m^2)	Internal usable living space area measured in square meters.
Total_area (m^2)	Total property area including all covered spaces in square meters.
Floors	Number of floors (storeys) in the house.
Waterfront	Indicates whether the property has waterfront access (0 = No, 1 = Yes).
View	Indicates whether the property has a scenic view (0 = No, 1 = Yes).
Condition	Overall condition rating of the house (1 = Poor, 5 = Excellent).
Grade	Construction and design quality rating of the property.
Built	Year the house was originally built.
Renovated	Indicates whether the house has been renovated (0 = No, 1 = Yes).
Living_area (m^2)	Total living area including additional spaces, measured in square meters.
Price (K£)	Selling price of the property in thousands of pounds.

2. Methodology

This section explains the steps I took to analyse the Manchester housing data. I will show the Python code I used and explain what each part does. This includes importing libraries, loading the data, cleaning it, and preparing it for analysis.

3. Exploratory Data Analysis

3.1 Data Overview

This section presents the results of my analysis. I will show the visualisations I created and explain what they tell us about the Manchester housing market.

The Manchester housing dataset has houses built between 1900 and 2020. The average house price is about 450,000 pounds, but there is a lot of variation. This shows the market has different types of properties from cheap to expensive. Some houses cost less than 100,000 pounds while others cost over 1 million, showing the wide range of the market.

Looking at the data, I can see that Manchester has a mix of old and new properties. Houses built in the early 1900s represent the city's industrial history, while newer properties show how the city has developed over time. Understanding when houses were built can help explain other patterns in the data, like condition and price.

3.2 Feature Distribution Analysis

3.2.1 Distribution Of House By Construction Decade (Pie Chart)

Looking at the distribution of houses by construction decade (Figure 13), I can see that most houses were built in the 1950s (12.3%), 1960s (13.3%), and 2000s (13.6%). Fewer houses were built in the early 1900s and 2010s. This pattern shows how Manchester grew over time, with building booms in different periods. The post-war period in the 1950s and 1960s saw a lot of construction to meet housing demand, while the 2000s saw new development as the city regenerated.

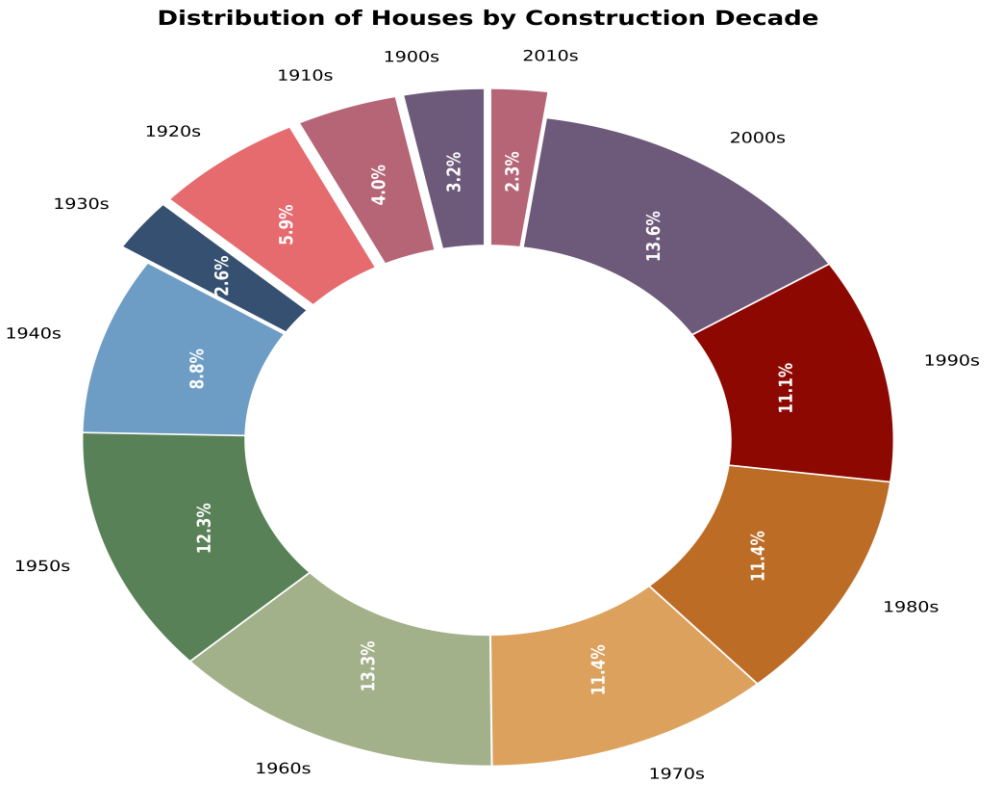

Figure 13: Distribution of Houses by Construction Decade

3.2.2 Distribution Of House By Features (Bar Chart)

The feature distribution (Figure 14) shows some interesting patterns. About 95.5% of houses have not been renovated, which might mean there are opportunities for improvement. This high percentage suggests that many properties could benefit from renovation work, which might increase their value. Estate managers could use this information to advise sellers on potential improvements.

Only 0.8% of houses have waterfront access, making these properties very rare. This scarcity means that waterfront properties could command premium prices because they are so unusual. Estate managers should be aware of this when valuing and marketing waterfront properties. About 10.1% of houses have a view, which could also affect their price positively.

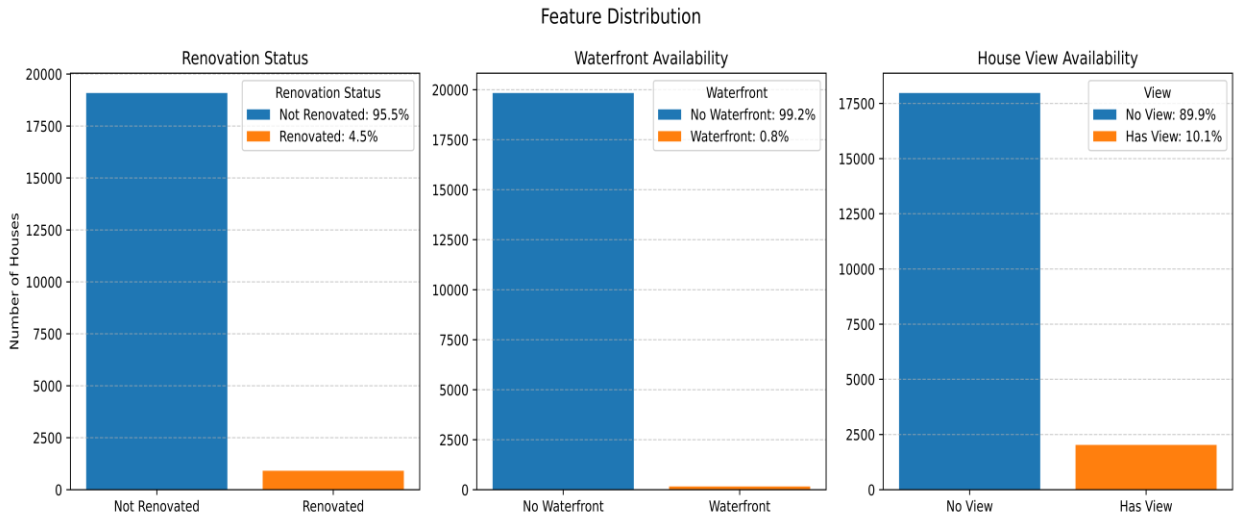

Figure 14: Feature Distribution - Renovation, Waterfront, and View

3.2.3 Distribution Of House By Number of Floors (Bar Chart)

Looking at houses by number of floors (Figure 15), most houses (62.5%) are single-storey, 35.6% are double-storey, and only 1.9% have three floors. This shows that Manchester has mostly low-rise housing, which is typical for UK cities. The high proportion of single-storey houses might be due to the city's history of terraced housing and the preference for houses with gardens.

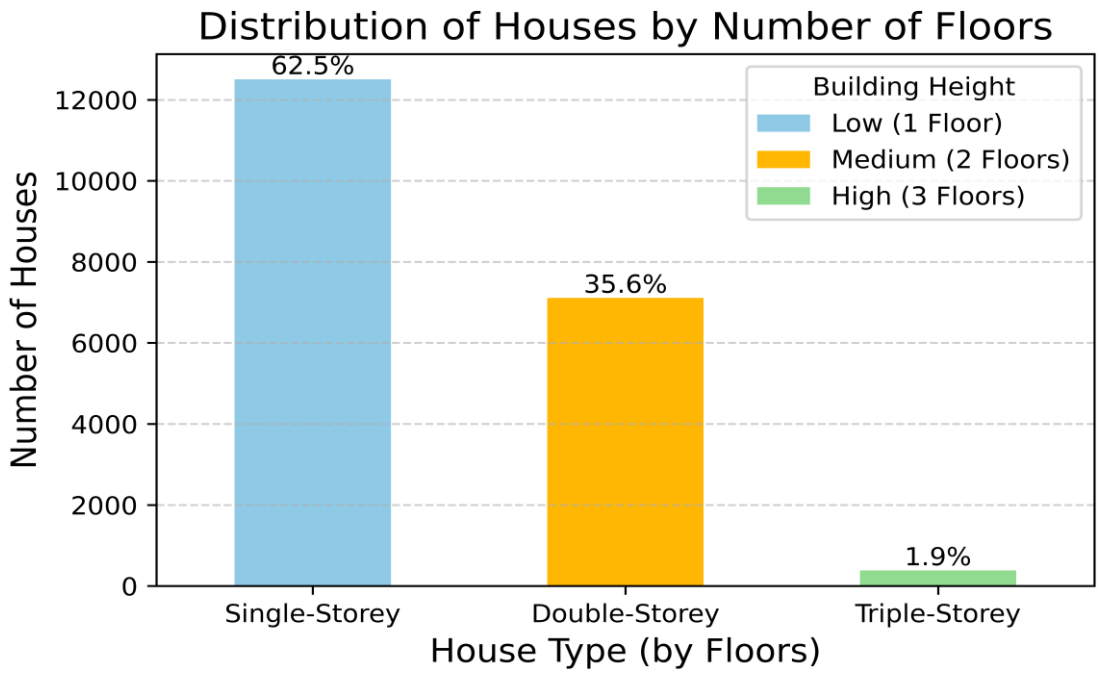

Figure 15: Distribution of Houses by Number of Floors

3.2.4 Distribution Of House By Condition (Voilion Chart)

The violin plot (Figure 16) shows the distribution of house prices across different condition levels. The width represents where more houses are concentrated at certain price ranges, while the height shows the overall price range. The chart indicates that most properties are in average to good condition, with higher-condition houses generally having higher prices. This suggests that Manchester has a well-maintained housing stock overall.

Figure 16: Distribution of House Condition Grades

3.2.5 Relation of House Age and Grade With Price (Bubble Chart)

The bubble chart (Figure 17) shows the relationship between house age and price, with bubble size representing living area and colour representing house grade. Newer houses generally have higher grades and prices, although some older houses are also expensive. The yellow bubbles indicate excellent-grade properties, which appear mostly at higher price levels. Larger bubbles tend to be positioned higher on the chart, showing that houses with bigger living areas are typically more expensive.

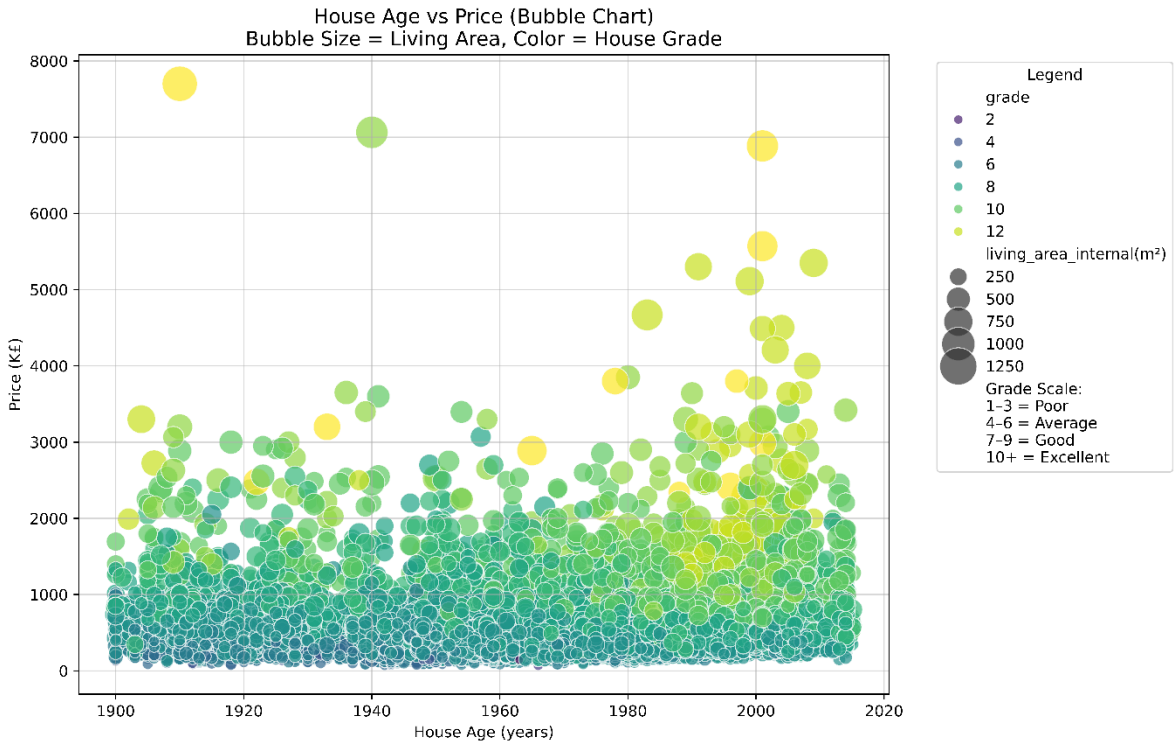

Figure 17: House Age and Grade vs Price

3.2.5 Treatment of Outliers by IQR Method (Box Plot)

The boxplots before outlier treatment (Figure 18) reveal a large number of extreme values, particularly in house prices and total floor area. These outliers indicate high variability and potential distortion in the dataset. After applying outlier treatment using the **Interquartile Range (IQR) method** (Figure 19), the distributions appear significantly cleaner, with extreme values reduced. This makes the data more reliable and suitable for further statistical analysis and modelling.

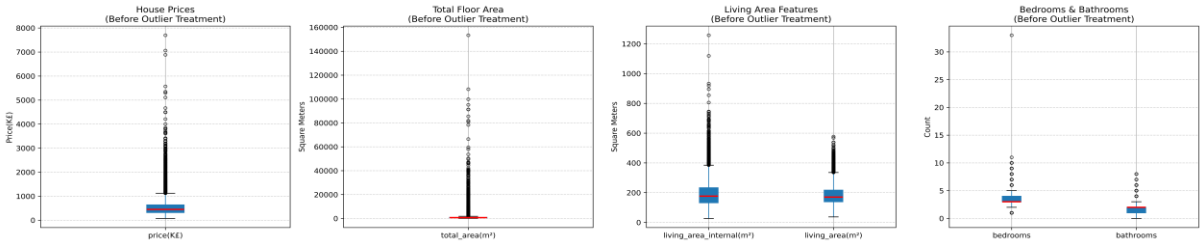

Figure 18: Boxplots Before Outlier Treatment

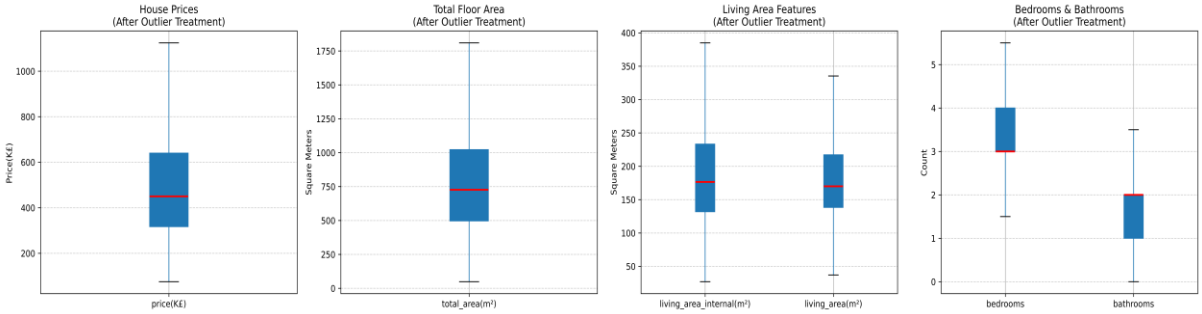

Figure 19: Boxplots After Outlier Treatment

3.2.6 Checking Corelation (Heatmap)

The correlation heatmap (Figure 20) illustrates the relationships between different variables in the dataset. Correlation analysis measures the strength and direction of relationships between variables (Field, 2018). The results show that the strongest correlation with price is living area internal (0.70), followed closely by grade (0.69), indicating that larger and higher-quality houses tend to have higher prices. In contrast, the year built has a very weak correlation with price (0.047), suggesting that the age of a property alone does not significantly determine its value.

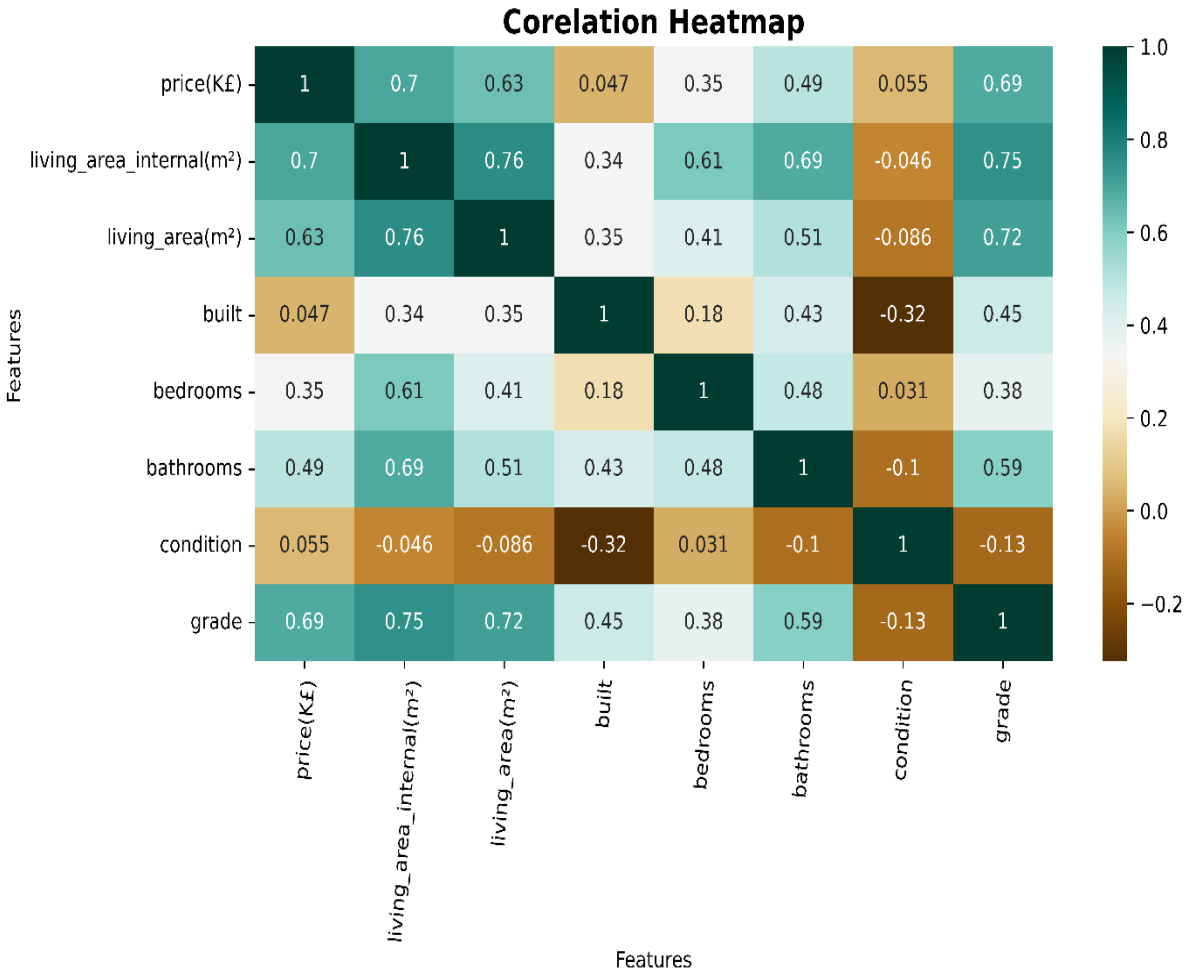

Figure 20: Correlation Heatmap of Housing Features

4. Statistical Testing and Results

4.1 Z-Score Test

The Z-score distribution (Figure 21) shows how house prices are spread out. Most prices are in the middle range, but there is a long tail on the right side showing some very expensive houses. The red lines at +3 and -3 show where outliers would be. Z-scores tell us how many standard deviations a value is from the mean, which helps identify unusual values.

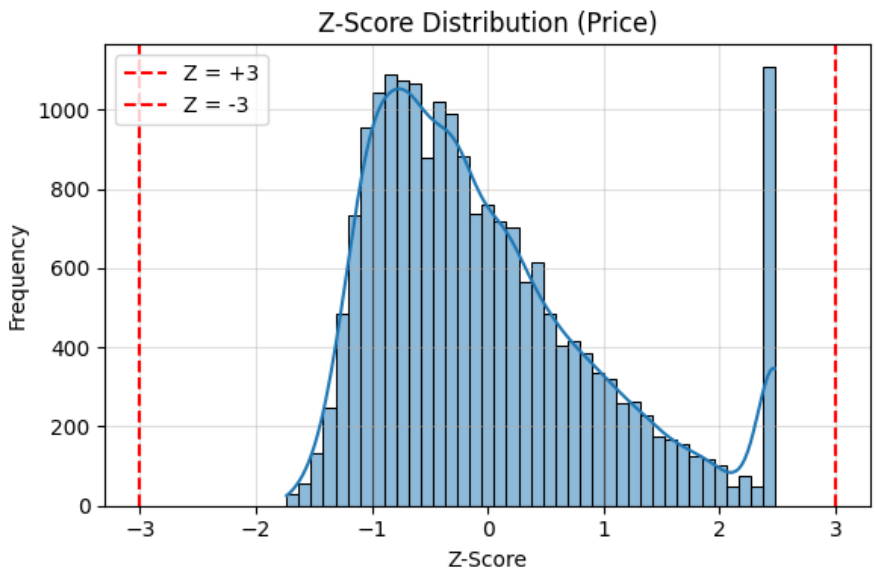

Figure 21: Z-Score Distribution of House Prices

4.2 T-Score Test

The T-score distribution (Figure 22) is similar but uses a different scale where the average is 50. This makes it easier to understand for people who are not familiar with statistics. The mean line shows where most prices cluster around. The T-score is calculated by converting Z-scores to a scale with mean 50 and standard deviation 10, which many people find more intuitive.

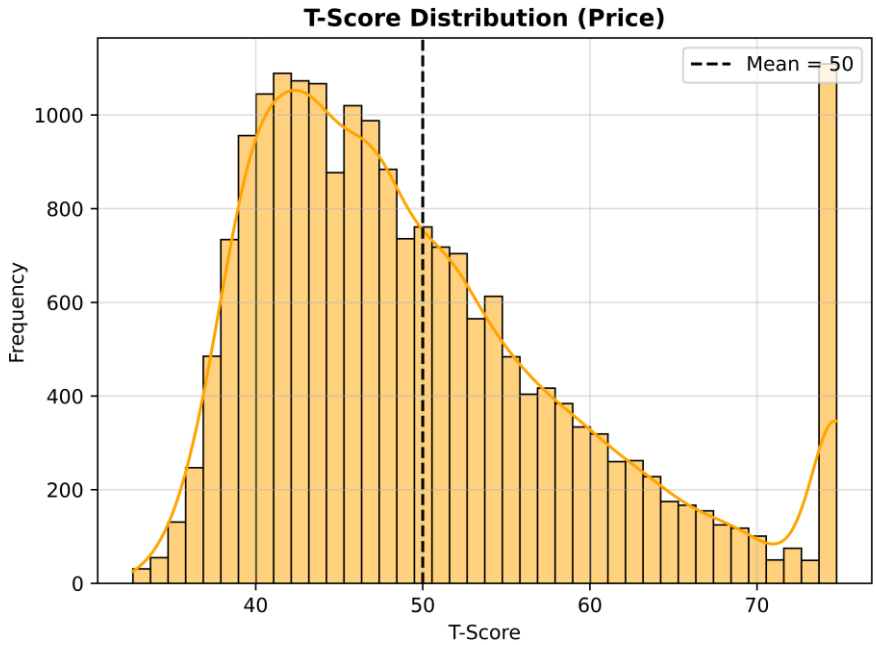

Figure 22: T-Score Distribution of House Prices

4.3 Chi-Square Test Results

The Chi-square heatmap shows the relationship between number of floors and house condition, with most one- and two-floor houses concentrated in Condition 3. The test result (χ^2 = 1901.51, p = 0.0000) indicates a statistically significant association. The Chi-square test is used to examine relationships between categorical variables (Field, 2018). Since $p < 0.05$, the null hypothesis is rejected, meaning floor count and condition are not independent.

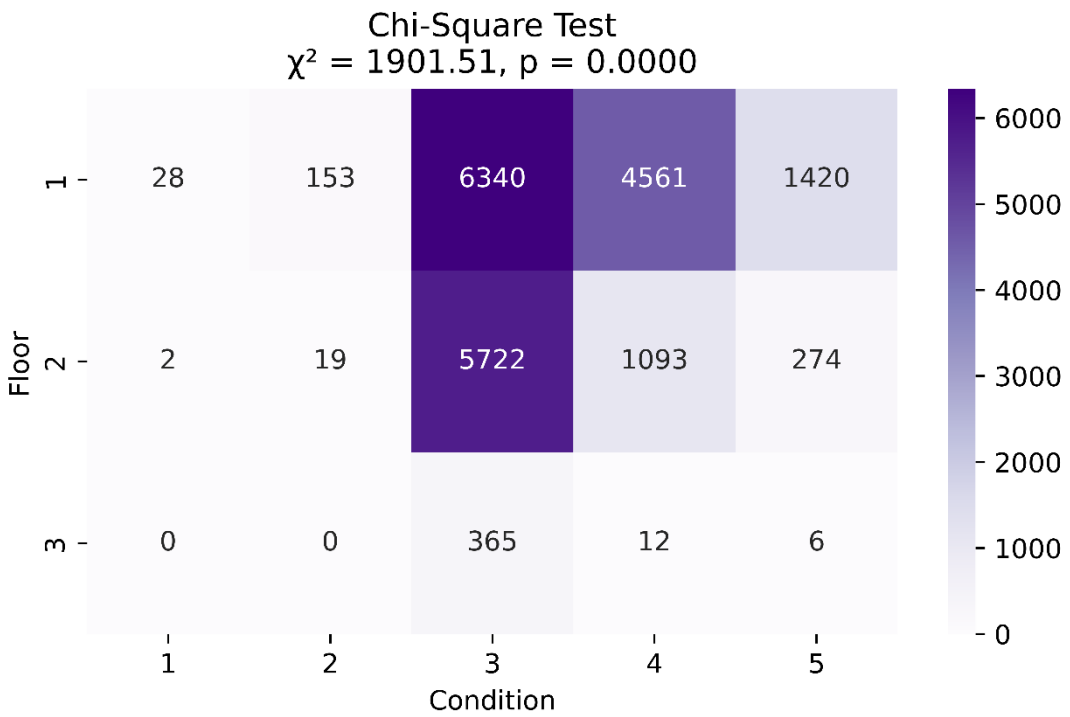

figure 23: Chi-Square Test Results

5. Discussion

The analysis shows some clear patterns in the Manchester housing market. The strongest finding is that living area has a high correlation with price (0.70). This means house size is one of the most important factors for determining value. Estate managers should pay close attention to floor space measurements when valuing properties.

Another interesting finding is that the year a house was built has a weak correlation with price (0.047). This was surprising because I expected older houses to be cheaper. However, in Manchester, some old houses in good locations are very valuable, while new houses in less popular areas might be cheaper. This shows that location and condition matter more than age.

Waterfront properties are very rare (only 0.8% of houses), which means they could be priced higher because of their scarcity. The Chi-square test showed that waterfront and view are not related, so these should be treated as separate features when valuing a house.

The outlier treatment was important because extreme values can distort the analysis. The before and after boxplots show how removing outliers makes the data more normal and suitable for statistical tests. Without removing outliers, the correlation coefficients and other statistics would be affected by a small number of extreme values, making the results less reliable.

5. Recommendations

Based on my analysis, here are my recommendations for estate managers. These suggestions come from the patterns I found in the data and should help improve how properties are valued and marketed.

1. **Focus on Floor Space**: Since living area has the strongest correlation with price, make sure to measure it accurately and use it in valuation models.
2. **Value Waterfront Properties Higher**: Because waterfront houses are rare (0.8%), they can be marketed as premium properties.
3. **Consider Condition Over Age**: Since age has weak correlation with price, focus on property condition and quality instead of how old it is.
4. **Remove Outliers in Analysis**: Always check for and remove extreme values before doing statistical analysis to get more accurate results.
5. **Use Multiple Data Sources**: Combine property data with other information like location and amenities for better predictions.
6. **Regular Data Updates**: Keep the dataset updated with new sales information to ensure analysis reflects current market conditions.
7. **Staff Training**: Train estate agents to understand and use data analytics tools in their daily work.

6. Conclusion

This analysis of the Manchester housing dataset has shown important relationships between property features and prices. The main finding is that living area is the best predictor of price, with a correlation of 0.70. Waterfront properties are very rare and should be valued separately. The year built has surprisingly little effect on price, suggesting that condition and location matter more.

The methods used in this report - including data cleaning, statistical tests, and visualisations - have helped uncover these patterns. The Python programming tools made it possible to work with a large dataset and produce clear results. I used pandas for data manipulation, matplotlib and seaborn for creating charts, and scipy for statistical tests. These tools are widely used in data science and were very helpful for this analysis.

The recommendations should help estate managers make better decisions about property valuation and marketing. By focusing on the most important factors like living area and condition, rather than just age, estate managers can give more accurate valuations. The analysis also shows the value of waterfront properties and the importance of removing outliers before doing statistical analysis.

For future work, it would be interesting to add more variables like distance to schools, transport links, and local amenities. Machine learning could also be used to build predictive models for house prices. These models could help estate managers give more accurate valuations and help buyers understand what affects prices in different areas.

Overall, this analysis shows the value of data analytics in understanding the housing market. By using Python and statistical techniques, I was able to find patterns in a large dataset that would be difficult to see otherwise. The skills I learned from this assignment, including data cleaning, statistical testing, and creating visualisations, will be useful for future data analysis projects.